14.	$P(k) \Rightarrow \sim (Q(y) \wedge L(k,y))$	13, E18
15.	$\sim (Q(y) \wedge L(k,y))$	4,14, I3
16.	$\sim Q(y) \vee \sim L(k,y)$	15, E16
17.	$\sim L(k,y) \vee \sim Q(y)$	16, E10
18.	$L(k,y) \Rightarrow \sim Q(y)$	17, E18
19.	$D(y) \Rightarrow \sim Q(y)$	7,18, I5
20.	$\sim D(y) \vee \sim Q(y)$	19, E18
21.	$\sim (D(y) \wedge Q(y))$	20, E16
22.	$(\forall x) \sim (D(x) \wedge Q(x))$	21, Gen
23.	$\sim (\exists x)(D(x) \wedge Q(x))$	22, FE8

Hence, P1, P2 $\vdash$ C ■

Example 9. [Lewis Carroll] There are no pencils of mine in this box. No sugar plums of mine are cigars. The whole of my property, that is not in this box, consists of cigars.

Conclusion: No pencils of mine are sugar plums.

Solution: Let U(the Universe of discourse) be the set of all my things;

$P(x)$ denote x is a pencil;
$B(x)$ denote x is in this box;
$C(x)$ denote x is a cigar;
$S(x)$ denote x is a sugar plum;

Then, the corresponding FOL-wffs are as follows.

P1: $\sim (\exists x)(P(x) \wedge B(x))$
P2: $\sim (\exists x)(S(x) \wedge C(x))$
P3: $(\forall x)(\sim B(x) \Rightarrow C(x))$
C: $\sim (\exists x)(P(x) \wedge S(x))$

We shall prove that: P1, P2, P3 $\vdash$ C

1.	$\sim (\exists x)(P(x) \wedge B(x))$	from Γ
2.	$\sim (\exists x)(S(x) \wedge C(x))$	from Γ
3.	$(\forall x)(\sim B(x) \Rightarrow C(x))$	from Γ
4.	$(\forall x) \sim (P(x) \wedge B(x))$	1, FE8
5.	$(\forall x) \sim (S(x) \wedge C(x))$	2, FE8
6.	$\sim (P(x) \wedge B(x))$	4, UI
7.	$\sim (S(x) \wedge C(x))$	5, UI
8.	$\sim P(x) \vee \sim B(x)$	6, E16
9.	$\sim S(x) \vee \sim C(x)$	7, E16
10.	$P(x) \Rightarrow \sim B(x)$	8, E18
11.	$S(x) \Rightarrow \sim C(x)$	9, E18
12.	$\sim B(x) \Rightarrow C(x)$	3, UI
13.	$P(x) \Rightarrow C(x)$	10,12, I5
14.	$\sim\sim C(x) \Rightarrow \sim S(x)$	11, E19
15.	$C(x) \Rightarrow \sim S(x)$	14, E15
16.	$P(x) \Rightarrow \sim S(x)$	13,15, I5
17.	$\sim P(x) \vee \sim S(x)$	16, E18

18. $\sim (P(x) \wedge S(x))$	17, E16
19. $(\forall x) \sim (P(x) \wedge S(x))$	18, Gen
20. $\sim (\exists x)(P(x) \wedge S(x))$	19, FE8

Hence, P1, P2, P3 $\vdash$ C ■

Example 10. [Lewis Carroll] When I work on a logic example without grumbling, you may be sure that it is one that I understand. This problem is not arranged in proper order, like the one's I am used to. No easy problem gives me a headache. I can't understand problems that aren't arranged in proper order, like those I am used to. I never grumble at a problem unless it gives me a headache.

Conclusion: This problem is difficult.

Solution: Let U(the Universe of discourse) be the set of logic problems I am working on;

$G(x)$ denote x makes me grumbling;
$U(x)$ denote I understand x;
$A(x)$ denote x is arranged in proper order like those I am used to;
$D(x)$ denote x is difficult;
$H(x)$ denote x gives me headache;
c denote this problem. (because this problem refers to a particular problem in U)

Then, the corresponding FOL-wffs are as follows.

P1: $(\forall x)(\sim G(x) \Rightarrow U(x))$
P2: $\sim A(c)$
P3: $\sim (\exists x)(\sim D(x) \wedge H(x))$
P4: $(\forall x)(\sim A(x) \Rightarrow \sim U(x))$
P5: $(\forall x)(G(x) \Rightarrow H(x))$ (Note: the sentence corresponds to P5 is of the form '*not p* unless *q*')
C: $D(c)$

We shall prove that: P1, P2, P3, P4, P5 $\vdash$ C.

1.	$(\forall x)(\sim A(x) \Rightarrow \sim U(x))$	from Γ
2.	$\sim A(c) \Rightarrow \sim U(c)$	1, UI
3.	$\sim A(c)$	from Γ
4.	$\sim U(c)$	3,2,I3
5.	$(\forall x)(\sim G(x) \Rightarrow U(x))$	from Γ
6.	$\sim G(c) \Rightarrow U(c)$	5, UI
7.	$\sim U(c) \Rightarrow \sim\sim G(c)$	6,E19
8.	$\sim U(c) \Rightarrow G(c)$	7,E15
9.	$G(c)$	4,8,I3
10.	$(\forall x)(G(x) \Rightarrow H(x))$	from Γ
11.	$G(c) \Rightarrow H(c)$	10, UI
12.	$H(c)$	9,11,I3
13.	$\sim (\exists x)(\sim D(x) \wedge H(x))$	from Γ
14.	$(\forall x) \sim (\sim D(x) \wedge H(x))$	13, FE8
15.	$(\forall x)(\sim\sim D(x) \vee \sim H(x))$	14, E16
16.	$(\forall x)(\sim D(x) \Rightarrow \sim H(x))$	15, E18
17.	$\sim D(c) \Rightarrow \sim H(c)$	16, UI
18.	$(H(c) \Rightarrow D(c))$	17,E19

19. $D(c)$ 12,18,I3

Hence, P1, P2, P3, P4, P5 $\vdash$ C ■

An alternative solution:

Let U(the Universe of discourse) be the set of logic problems I am working on;

$G(x)$ denote x makes me grumbling;
$U(x)$ denote I understand x;
$A(x)$ denote x is arranged in proper order like those I am used to;
$D(x)$ denote x is difficult;
$H(x)$ denote x gives me headache;
$T(x)$ denote x is this problem.

Then, the corresponding FOL-wffs are as follows.

P1: $(\forall x)(\sim G(x) \Rightarrow U(x))$
P2: $\sim (\exists x)(T(x) \wedge A(x))$
P3: $\sim (\exists x)(\sim D(x) \wedge H(x))$
P4: $(\forall x)(\sim A(x) \Rightarrow \sim U(x))$
P5: $(\forall x)(G(x) \Rightarrow H(x))$
C: $(\forall x)(T(x) \Rightarrow D(x))$

We shall prove that: P1, P2, P3.P4, P5 $\vdash$ C.

1.	$(\forall x)(\sim A(x) \Rightarrow \sim U(x))$	from Γ
2.	$\sim A(x) \Rightarrow \sim U(x)$	1, UI
3.	$\sim (\exists x)(T(x) \wedge A(x))$	from Γ
4.	$(\forall x) \sim (T(x) \wedge A(x))$	3,FE8
5.	$(\forall x)(\sim T(x) \vee \sim A(x))$	4,E16
6.	$(\forall x)(T(x) \Rightarrow \sim A(x))$	5,E18
7.	$(T(x) \Rightarrow \sim A(x))$	6,UI
8.	$(T(x) \Rightarrow \sim U(x))$	7,2,I5
9.	$(\forall x)(\sim G(x) \Rightarrow U(x))$	from Γ
10.	$\sim G(x) \Rightarrow U(x)$	9, UI
11.	$\sim U(x) \Rightarrow \sim\sim G(x)$	10,E19
12.	$\sim U(x) \Rightarrow G(x)$	11,E15
13.	$T(x) \Rightarrow G(x)$	8,12,I5
14.	$(\forall x)(G(x) \Rightarrow H(x))$	from Γ
15.	$G(x) \Rightarrow H(x)$	9, UI
16.	$T(x) \Rightarrow H(x)$	13,15,I5
17.	$\sim (\exists x)(\sim D(x) \wedge H(x))$	from Γ
18.	$(\forall x) \sim (\sim D(x) \wedge H(x))$	17, FE8
19.	$(\forall x)(\sim\sim D(x) \vee \sim H(x))$	18, E16
20.	$(\forall x)(\sim D(x) \Rightarrow \sim H(x))$	19, E18
21.	$(\sim D(x) \Rightarrow \sim H(x))$	20, UI
22.	$(H(x) \Rightarrow D(x))$	21,E19
23.	$T(x) \Rightarrow D(x)$	16,22,I5
24.	$(\forall x)(T(x) \Rightarrow D(x))$	23,Gen

Hence, $\vdash (\mathrm{P1} \wedge \mathrm{P2} \wedge \mathrm{P3} \wedge \mathrm{P4} \wedge \mathrm{P5}) \Rightarrow \mathrm{C}$ ■

Example 11. Every student passes at least one of the courses he takes. Some students dislike every course they take.

Conclusion: Some students pass some couses they dislike.

Solution: Let U(the Universe of discourse) be the set of all things;

$S(x)$ denote x is a student;
$C(x)$ denote x is a course;
$T(x,y)$ denote x takes course y;
$P(x,y)$ denote x passes course y;
$D(x,y)$ denote x dislikes y;

Then, the corresponding FOL-wffs are as follows.

P1: $(\forall x)(S(x) \Rightarrow (\exists y)(C(y) \wedge T(x,y) \wedge P(x,y)))$
P2: $(\exists x)(S(x) \wedge (\forall y)(C(y) \wedge T(x,y) \Rightarrow D(x,y)))$
C: $(\exists x)(\exists y)(S(x) \wedge C(y) \wedge T(x,y) \wedge P(x,y) \wedge D(x,y))$

We shall prove that: $\mathrm{P1}, \mathrm{P2} \vdash \mathrm{C}$.

(*Direct proof*)

1.	$(\forall x)(S(x) \Rightarrow (\exists y)(C(y) \wedge T(x,y) \wedge P(x,y)))$	from Γ
2.	$(\exists x)(S(x) \wedge (\forall y)(C(y) \wedge T(x,y) \Rightarrow D(x,y)))$	from Γ
3.	$S(k) \wedge (\forall y)(C(y) \wedge T(k,y) \Rightarrow D(k,y))$	2, EI, k is a constant
4.	$S(k) \Rightarrow (\exists y)(C(y) \wedge T(k,y) \wedge P(k,y))$	1, UI
5.	$S(k)$	3, I2
6.	$(\exists y)(C(y) \wedge T(k,y) \wedge P(k,y))$	5,4, I3
7.	$(C(c) \wedge T(k,c) \wedge P(k,c))$	6, EI,c is a (new) constant
8.	$(\forall y)(C(y) \wedge T(k,y) \Rightarrow D(k,y))$	3, E9,I2
9.	$C(c) \wedge T(k,c) \Rightarrow D(k,c)$	6, UI
10.	$C(c) \wedge T(k,c)$	7, I2
11.	$D(k,c)$	10,9, I3
12.	$S(k) \wedge (C(c) \wedge T(k,c) \wedge P(k,c))$	5,7, I6
13.	$(S(k) \wedge (C(c) \wedge T(k,c))) \wedge P(k,c)$	12, E11
14.	$((S(k) \wedge C(c)) \wedge T(k,c)) \wedge P(k,c)$	13, E11
15.	$(S(k) \wedge C(c) \wedge T(k,c) \wedge P(k,c)) \wedge D(k,c)$	14,11, I6
16.	$(\exists y)(S(k) \wedge C(y) \wedge T(k,y) \wedge P(k,y) \wedge D(k,y))$	15, EQ
17.	$(\exists x)(\exists y)(S(x) \wedge C(y) \wedge T(x,y) \wedge P(x,y) \wedge D(x,y))$	16, EQ

Hence, $\mathrm{P1}, \mathrm{P2} \vdash \mathrm{C}$ ■

2.10 First-order-logic with Equality

First, we shall generalize the arguments of predicates from variables to expressions which we call terms.

Definition: A *term* is:
(a) a variable or a constant, or

(b) $f(t_1, t_2, \cdots, t_n)$, in which f is a letter representing an n-argument function and $t_1, t_2, \cdots, t_n$ are terms.

(c) No expression other than those which can be defined by means of a finite number of applications of (a) and (b) is a term.

Example: Let $add(x, y)$ denote the *addition* function, i.e. $add(x, y) = x + y$, and $exp(x, y)$ denote the *exponentiation* function. i.e. $exp(x, y) = x^y$. Both functions are 2–argument functions. Then $exp(add(add(x, y), 1), 3)$ is a term. It denotes the mathematical expression $((x + y) + 1)^3$. ■

We shall also extend the notion of free and bound variables to terms.

Definition: Let t be a term. An occurrence of t in a FOL-wff α is ***free*** in α if no occurrence of a variable in t is bound in α. The occurrence of t is ***bound*** in α, otherwise. Let $\alpha(s)$ be a FOL-wff containing some occurrences of a term s. $\alpha(s)$ is ***free for*** a term t if no occurrence of t which replaces a free occurrence of s is bound in $\alpha(t)$, where $\alpha(t)$ is the FOL-wff resulting from $\alpha(s)$ after some (not necessarily all) free occurrences of s are replaced by t.

Example: The FOL-wff $(\exists y)(Greater(x, y))$ is not free for the term $add(y, 1)$. This is because the free occurrence of x in it is within the scope of $(\exists y)$. Therefore, if we substitute $add(y, 1)$ for x in $(\exists y)(Greater(x, y))$, the free y in $add(y, 1)$ becomes bound in $(\exists y)(Greater(add(y, 1), y))$. $(\exists y)(Greater(x, y))$ is free for the term $add(z, 1)$, however. ■

Remark: It can be shown that the results we have obtained for first-order logic remain valid even if we allow terms to be used as arguments in predicates. In our subsequent discussions that involve first-order logic, we shall assume that terms (including variables and constants) are used as arguments in predicates.

Definition: An ***alphabet*** is a collection of symbols. A ***sentence*** is a finite sequence of symbols of an alphabet. A ***language*** is a collection of sentences made out of symbols of an alphabet. A ***theory*** consists of a language containing a pre-designated collection of sentences called axioms, and a collection of rules for manipulating sentences in the language.

Example 2.10.9 The first-order-logic is a theory in which the language consists of terms and FOL-wffs, the axioms are the FOL-wffs of the form $(\omega \vee \sim \omega)$, and the rules are the equivalences, the inference rules and the rules governing the applications of equivalences and inference rules.

Definition: A ***first-order theory*** is a first-order-logic with a specific collection of constant, function and predicate symbols, agumented with a collection (possibly empty) of axioms called ***proper axioms***.

Definition: A ***first-order theory with equality*** is a first-order theory which includes a two-argument predicate, $E(x, y)$, customarily written as $x = y$, for which the following axioms hold.

A1. (*reflexivity*) $\vdash (\forall x)x = x$

A2. (*symmetricity*) $\vdash (\forall x)(\forall y)(x = y \Rightarrow y = x)$

A3. (*transitivity*) $\vdash (\forall x)(\forall y)(\forall z)((x = y) \wedge (y = z) \Rightarrow x = z)$

A4. (*substitutivity of functions*)

For every n-argument function f^n and for each $i, 1 \leq i \leq n$,

$\vdash (\forall x)(\forall y)(\forall z_1) \cdots (\forall z_{i-1})(\forall z_{i+1}) \cdots (\forall z_n)$

$$(x = y \Rightarrow (f^n(z_1, \cdots z_{i-1}, x, z_{i+1}, \cdots, z_n) = f^n(z_1, \cdots z_{i-1}, y, z_{i+1}, \cdots, z_n)))$$

A5. (*substitutivity of predicates*)

For every n-argument predicate P^n (other than =) and for each $i, 1 \leq i \leq n$,

$\vdash (\forall x)(\forall y)(\forall z_1)\cdots(\forall z_{i-1})(\forall z_{i+1})\cdots(\forall z_n)$

$$(x = y \Rightarrow (P^n(z_1, \cdots z_{i-1}, x, z_{i+1}, \cdots, z_n) \Rightarrow P^n(z_1, \cdots z_{i-1}, y, z_{i+1}, \cdots, z_n)))$$

The notation $x \neq y$ denotes $\sim (x = y)$.

Since the symbol = denotes a predicate, it thus has higher priority then the logical operators and the quantifiers.

Example 2.10.10

(i) Let $\mathbb{G}$ be the first-order theory whose constant, function and predicate symbols are restricted to 0, f^2, and =, respectively, where $f^2(x, y)$ is abbreviated as $x+y$. The proper axioms of $\mathbb{G}$ are the following.

(a) $(\forall x)(\forall y)(\forall z)x + (y + z) = (x + y) + z$

(b) $(\forall x)x + 0 = x$

(c) $(\forall x)(\exists y)x + y = 0$

(d) $(\forall x)x = x$

(e) $(\forall x)(\forall y)(x = y \Rightarrow y = x)$

(f) $(\forall x)(\forall y)(\forall z)(x = y \Rightarrow (y = z \Rightarrow x = z))$

(g) $(\forall x)(\forall y)(\forall z)(x = y \Rightarrow (x + z = y + z \wedge z + x = z + y))$

We shall prove that $\mathbb{G}$ is a first-order theory with equality.

First, note that A1 and A2 follow immediately from Axioms (d) and (e), respectively. A3 is a direct consequence of Axiom (f) and Theorem 1.7.2(v).

The substitutivity of the function f^2 (or equivalently, +) follows from Axiom (g) as follows.

1. $(\forall x)(\forall y)(\forall z)(x = y \Rightarrow (x + z = y + z \wedge z + x = z + y))$ Axiom (g)
2. $(x = y \Rightarrow (x + z = y + z \wedge z + x = z + y))$ 1, UI(*three times*)
3. $\sim (x = y) \vee (x + z = y + z \wedge z + x = z + y)$ 2, E18
4. $(\sim (x = y) \vee (x + z = y + z)) \wedge (\sim (x = y) \vee (z + x = z + y))$ 3, E14
5. $\sim (x = y) \vee (x + z = y + z)$ 4, I2
6. $\sim (x = y) \vee (z + x = z + y)$ 4, E9,I2
7. $(x = y) \Rightarrow (x + z = y + z)$ 5, E18
8. $(x = y) \Rightarrow (z + x = z + y)$ 6, E18
9. $(\forall x)(\forall y)(\forall z)(x = y) \Rightarrow (x + z = y + z)$ 7, Gen(*three times*)
10. $(\forall x)(\forall y)(\forall z)(x = y) \Rightarrow (z + x = z + y)$ 8, Gen(*three times*)

Note that lines 9 and 10 are the substitutivity property of the function f^2. ■

(ii) Let $\mathbb{F}$ be a first-order theory which has no constant and function symbols but two binary predicates, namely = and P^2, where $P^2(x, y)$ is abbreviated as $x \prec y$. The proper axioms of $\mathbb{F}$ are the following.

(a) $(\forall x)x = x$

(b) $(\forall x)(\forall y)(x = y \Rightarrow y = x)$

(c) $(\forall x)(\forall y)(\forall z)(x = y \Rightarrow (y = z \Rightarrow x = z))$

(d) $(\forall x)(\forall y)(\forall z)(x \prec y \wedge y \prec z \Rightarrow x \prec z)$

(e) $(\forall x)(\forall y)(x = y \Rightarrow \sim (x \prec y))$

(f) $(\forall x)(\forall y)(x \prec y \vee x = y \vee y \prec x)$

We shall prove that F is a first-order theory with equality.

First, note that A1 and A2 follow immediately from Axioms (a) and (b), respectively, and A3 is a direct consequence of Axiom (c) and Theorem 1.7.2(v).

It remains to check the substitutivity of the predicate P^2 (or equivalently, $\prec$).

First, we shall prove the left substitutivity property of $\prec$:

$$\vdash \quad (\forall x)(\forall y)(\forall z)(x = y \Rightarrow (x \prec z \Rightarrow y \prec z))$$

Proof:

1. $(\forall x)(\forall y)(\forall z)(x = y \Rightarrow (y = z \Rightarrow x = z))$ Axiom (c)
2. $x = y \Rightarrow (y = z \Rightarrow x = z)$ 1,UI(*three times*)
3. $(x = y) \wedge (y = z) \Rightarrow x = z$ 2,Theorem 1.7.2(v), substitution
4. $(\forall x)(\forall z)(x = z \Rightarrow \sim (x \prec z))$ Axiom (e)
5. $x = z \Rightarrow \sim (x \prec z)$ 4, UI(*twice*)
6. $(x = y) \wedge (y = z) \Rightarrow \sim (x \prec z)$ 3,5, I5
7. $x = y \Rightarrow (y = z \Rightarrow \sim (x \prec z))$ 6, Theorem 1.7.2(v), substitution
8. $x = y \Rightarrow ((x \prec z) \Rightarrow \sim (y = z))$ 7, E19,E15
9. $(x = y) \wedge (x \prec z) \Rightarrow \sim (y = z)$ 8, Theorem 1.7.2(v), substitution
10. $(\forall y)(\forall z)(y \prec z \vee y = z \vee z \prec y)$ Axiom (f)
11. $y \prec z \vee y = z \vee z \prec y$ 10, UI(*three times*)
12. $y = z \vee (z \prec y \vee y \prec z)$ 11, E10,E12
13. $\sim\sim (y = z) \vee (\sim\sim (z \prec y) \vee y \prec z)$ 12, E15
14. $\sim (y = z) \Rightarrow (\sim (z \prec y) \Rightarrow y \prec z)$ 13, E18(*twice*)
15. $(x = y) \wedge (x \prec z) \Rightarrow (\sim (z \prec y) \Rightarrow y \prec z)$ 9,14, I5
16. $((x = y) \wedge (x \prec z) \Rightarrow (\sim (z \prec y) \Rightarrow y \prec z)) \Rightarrow$
    $((x = y) \wedge (x \prec z) \Rightarrow \sim (z \prec y))) \Rightarrow ((x = y) \wedge (x \prec z) \Rightarrow (y \prec z)))$ Chapter 1, Ex. 2.d
17. $((x = y) \wedge (x \prec z) \Rightarrow \sim (z \prec y)) \Rightarrow ((x = y) \wedge (x \prec z) \Rightarrow (y \prec z))$ 15,16, I3
18. $(\forall x)(\forall y)(x = y \Rightarrow \sim (x \prec y))$ Axiom (e)
19. $x = y \Rightarrow \sim (x \prec y)$ 18, UI(*twice*)
20. $(\forall x)(\forall y)(\forall z)(x \prec z \wedge z \prec y \Rightarrow x \prec y)$ Axiom (e)
21. $(x \prec z \wedge z \prec y \Rightarrow x \prec y)$ 20, UI(*three times*)
22. $\sim (x \prec y) \Rightarrow \sim (x \prec z \wedge z \prec y)$ 21, E19
23. $x = y \Rightarrow \sim (x \prec z \wedge z \prec y)$ 19,22, I5
24. $x = y \Rightarrow (\sim (x \prec z) \vee \sim (z \prec y))$ 23, E16
25. $x = y \Rightarrow ((x \prec z) \Rightarrow \sim (z \prec y))$ 24, E18
26. $(x = y) \wedge (x \prec z) \Rightarrow \sim (z \prec y)$ 25,Theorem 1.7.2(v), substitution
27. $(x = y) \wedge (x \prec z) \Rightarrow (y \prec z)$ 26,17, I3
28. $(x = y) \Rightarrow ((x \prec z) \Rightarrow (y \prec z))$ 27,Theorem 1.7.2(v), substitution
29. $(\forall x)(\forall y)(\forall z)((x = y) \Rightarrow ((x \prec z) \Rightarrow (y \prec z)))$ 28, Gen(*three times*)

The right substitutivity of $\prec$, namely $\vdash (\forall x)(\forall y)(\forall z)(x = y \Rightarrow (z \prec x \Rightarrow z \prec y))$, can be proved similarly. ■

Axioms A4 and A5 infer the following substitutive property of equalities.

Theorem 2.10.4 *Let s, t be terms and $\alpha(s)$ be a FOL-wff which is free for t. Let $\alpha(t)$ be a FOL-wff resulting from $\alpha(s)$ after some (not necessarily all) free occurrences of s are replaced by t. Then,*

$$(s = t) \quad \Rightarrow \quad (\alpha(s) \Rightarrow \alpha(t))$$

Proof: (Omitted) ■

The above theorem is the basis of the following inference rule.

Sub$_=$ (*Subtitutivity of* =)

$$\frac{s = t,\ \alpha(s)}{\alpha(t)} \qquad (\alpha(s)\ is\ free\ for\ t)$$

where s and t are terms, and $\alpha(t)$ is obtained from $\alpha(s)$ by replacing some (not necessarily all) free occurrences of s by t.

Example 2.10.11 $\vdash (\forall x)(\exists y)x = y$.

Solution:

1. $(\forall x)x = x$ Axiom A1
2. $x = x$ 1,UI
3. $(\exists y)x = y$ 2, EQ
4. $(\forall x)(\exists y)x = y$ 3, Gen

Hence, $\vdash (\forall x)(\exists y)x = y$. ■

With equality, it is possible to express not only the existence of an object having a specific property, but a unique object having a specific property. Specifically, the statement: "there exists *one and only one* x such that $\alpha(x)$ is true" can be expressed as:

$$(\exists x)\alpha(x) \ \wedge \ (\forall x)(\forall y)(\alpha(x) \wedge \alpha(y) \Rightarrow x = y)$$

The above FOL-wff is abbreviated as $(\exists!x)\alpha(x)$, read as "there exists a unique x for which $\alpha(x)$ is true." As a result, to prove that $(\exists!x)\alpha(x)$, we may divide the proof into two parts. In the first part, we prove $\vdash (\exists x)\alpha(x)$. In the second part, we prove $\vdash (\forall x)(\forall y)(\alpha(x) \wedge \alpha(y) \Rightarrow x = y)$. Then, by combining the results of the two parts, we obtain $\vdash (\exists x)\alpha(x) \wedge (\forall x)(\forall y)(\alpha(x) \wedge \alpha(y) \Rightarrow x = y)$ which is $(\exists!x)\alpha(x)$.

Example 2.10.12 $\vdash (\forall x)(\exists!y)(x = y)$.

Solution:

First, we shall prove that: $\vdash (\forall x)(\exists y)(x = y)$ . . . (I)

This has been done in Example 2.10.11.

Next, we shall prove that $\vdash (\forall x)(\forall y)(\forall z)((x = y \wedge x = z) \Rightarrow y = z)$.

1. $(\forall y)(\forall x)(\forall z)((y = x \wedge x = z) \Rightarrow y = z)$ Axiom A3
2. $(y = x \wedge x = z) \Rightarrow y = z$ 1, UI(*three times*)
3. $(\forall x)(\forall y)(x = y \Rightarrow y = x)$ Axiom A2
4. $x = y \Rightarrow y = x$ 3, UI(*twice*)
5. $y = x \Rightarrow (x = z \Rightarrow y = z)$ 2, Theorem 1.7.2(v), substitution

6. $x = y \Rightarrow (x = z \Rightarrow y = z)$ 4,5, I5
7. $(x = y \wedge x = z) \Rightarrow y = z$ 6, Theorem 1.7.2(v), substitution
8. $(\forall x)(\forall y)(\forall z)((x = y \wedge x = z) \Rightarrow y = z)$ 7, Gen(*three times*)

Therefore, $\vdash (\forall x)(\forall y)(\forall z)((x = y \wedge x = z) \Rightarrow y = z)$ . . . (II)

Now,

1. $(\forall x)(\exists y)(x = y)$ Example 2.10.11.
2. $(\forall x)(\forall y)(\forall z)((x = y \wedge x = z) \Rightarrow y = z)$ (II)
3. $(\forall x)(\exists y)(x = y) \wedge (\forall x)(\forall y)(\forall z)((x = y \wedge x = z) \Rightarrow y = z)$ 1,2, I6
4. $(\forall x)((\exists y)(x = y) \wedge (\forall y)(\forall z)((x = y \wedge x = z) \Rightarrow y = z))$ 3, FE9

Hence, $\vdash (\forall x)((\exists y)(x = y) \wedge (\forall y)(\forall z)((x = y \wedge x = z) \Rightarrow y = z))$, or equivalently, $\vdash (\forall x)(\exists!y)(x = y)$. ■

Theorems 1.7.1 and 2.7.3 show that both propositional logic and first-order logic are sound and complete. A question naturally arises is: "Is every first-order theory sound and complete?"

Arithmetic is a branch of mathematics that is most immediately intuitive. Therefore, it is not surprising that attempts to establish a rigorous foundation for mathematics started from arithmetic. The Peano axioms were introduced for this purpose. Using these axioms, a first-order theory for arithmetic was built.

Godel's Incompleteness Theorem: *Any mathematical system containing all of the theorems of arithmetic is incomplete.*

Proof: (Omitted) ■

Specifically, the theorem states that any mathematical system that is powerful enough to contain all of the rules (definitions, axioms, and theorems) of arithmetic must contain some statement that is valid but for which no formal proof can ever be derived based on that system.

2.11 Exercises

2.1. Given the following interpretation:
Universe of Discourse: U = {1,2};
Assignment of constants: $a = 2$, $b = 1$;
Definition of function P:

$P(1)$	$P(2)$
T	F

Definition of predicate Q:

$Q(1,1)$	$Q(1,2)$	$Q(2,1)$	$Q(2,2)$
F	T	F	T

Evaluate the following well-formed formulas:

(a) $(\forall x)(\forall y)(P(x) \Rightarrow Q(x, y))$
(b) $(\exists x)(\forall y)(P(x) \Rightarrow Q(x, y))$
(c) $(\forall x)(\exists y)(P(x) \Rightarrow Q(x, y))$
(d) $(\exists x)(\exists y)(P(x) \Rightarrow Q(x, y))$
(e) $(\exists y) \sim Q(b, y)$

(f) $(\exists x)(Q(x,b) \Rightarrow P(b))$
(g) $(\exists x)(P(x) \Rightarrow (\forall y)(\exists x)(Q(y,x) \wedge P(x)))$
(h) $(\forall x)(\forall y)(Q(x,y) \Rightarrow (\exists z)(P(x) \vee Q(y,z)))$
(i) $(\exists x)(\exists y)(Q(x,a) \Rightarrow Q(b,y))$
(j) $(\exists x)((\forall y)Q(x,a) \wedge (\exists z)(P(z) \Rightarrow Q(b,y)))$
(k) $(\forall x)(\forall y)(Q(x,y) \Rightarrow (\sim P(y) \wedge (\exists z)(Q(y,z) \wedge \sim P(z))))$
(l) $(\exists x)((\forall y) \sim Q(y,x) \Rightarrow (P(x) \wedge (\exists z)(Q(z,x) \wedge Q(z,z))))$

2.2. Prove the following theorems ($\alpha(x)$, $\beta(x)$ etc. represent FOL wffs.):

(a) $\vdash (\exists x)(\forall y)\alpha(x,y) \Rightarrow (\forall y)(\exists x)\alpha(x,y)$.
(b) $\vdash (\exists x)(\exists y)\alpha(x,y) \Rightarrow (\exists y)(\exists x)\alpha(x,y)$.
(c) $\vdash (\forall x)\alpha(x) \vee (\forall x)\beta(x) \Rightarrow (\forall x)(\alpha(x) \vee \beta(x))$.
(d) $\vdash (\forall x)(\alpha(x) \Rightarrow \beta(x)) \Rightarrow ((\forall x) \sim \beta(x) \Rightarrow (\forall x) \sim \alpha(x))$.
(e) $\vdash (\forall x)(\alpha(x) \Rightarrow \beta) \Leftrightarrow ((\exists x)\alpha(x) \Rightarrow \beta)$ where x is not free in β.
(f) $\vdash (\alpha \Rightarrow (\forall x)\beta(x)) \Leftrightarrow (\forall x)(\alpha \Rightarrow \beta(x))$ where x is not free in α.
(g) $\vdash (\forall x)(\alpha(x) \vee \beta(x)) \Rightarrow (\forall x)\alpha(x) \vee (\exists x)\beta(x)$.
(h) $\vdash (\forall x)\beta(x) \Leftrightarrow (\forall x)(\sim\alpha(x) \Rightarrow \beta(x)) \wedge (\forall x)(\alpha(x) \Rightarrow \beta(x))$
(i) $\vdash (\forall x)\alpha(x,x) \Rightarrow (\forall x)(\exists y)\alpha(x,y)$
(j) $(\forall x)(\alpha(x) \Leftrightarrow (\exists y)\beta(y,x)), (\forall x)(\forall y)(\beta(x,y) \Rightarrow (\gamma(x) \wedge \gamma(y))) \vdash (\forall x)(\alpha(x) \Rightarrow \gamma(x))$
(k) $\vdash (\forall x)(\exists y)(\exists z)(\alpha(x,y,z) \vee \sim(\exists z)(\exists u) \sim\alpha(x,z,u)) \Rightarrow (\exists x)(\exists y)\alpha(a,y,x)$, where a is a constant.
(l) $\vdash (\forall x)(\forall y)\beta(x,y) \Leftrightarrow (\forall x)(\forall y)(\sim\alpha(x,y) \Rightarrow \beta(x,y)) \wedge (\forall x)(\forall y)(\alpha(x,y) \Rightarrow \beta(x,y))$

2.3. Prove that each of the following conclusions (C) follows from the given premises.

a. P1: $(\exists x)(\forall y)(P(x) \wedge (Q(y) \Rightarrow R(x,y)))$
P2: $(\forall x)(\forall y)((P(x) \wedge S(y)) \Rightarrow \sim R(x,y))$

C :$(\forall x)(Q(x) \Rightarrow \sim S(x))$

b. P1: $(\forall x)(P(x) \Rightarrow Q(x))$
P2: $(\forall x)(S(x) \Rightarrow R(x))$
P3: $\sim (\exists x)(Q(x) \wedge R(x))$

C :$(\forall x)(S(x) \Rightarrow \sim P(x))$

c. P1: $(\forall x)(\sim Q(x) \Rightarrow P(x))$
P2: $\sim (\exists x)(R(x) \wedge Q(x))$
P3: $\sim (\exists x)(S(x) \wedge P(x))$

C :$\sim (\exists x)(R(x) \wedge S(x))$

d. P1: $(\exists x)(P(x) \wedge (\forall y)((R(y) \wedge S(x,y)) \Rightarrow Z(x,y)))$
P2: $(\forall x)(P(x) \Rightarrow (\exists y)(R(y) \wedge \sim U(x,y) \wedge T(x,y)))$
P3: $(\forall x)(\forall y)((P(x) \wedge R(y) \wedge T(x,y)) \Rightarrow S(x,y))$

C :$(\exists x)(\exists y)(P(x) \wedge R(y) \wedge Z(x,y) \wedge \sim U(x,y))$

e. P1: $(\forall x)(P(x) \wedge (\exists y)(T(y) \wedge L(x,y)) \Rightarrow (\exists z)(\exists w)(D(z) \wedge T(w) \wedge L(x,w) \wedge L(z,w) \wedge B(x,z)))$
P2: $\sim (\exists x)(\exists y)(D(x) \wedge T(y) \wedge L(x,y))$

C :$\sim (\exists z)(\exists w)(P(z) \wedge T(w) \wedge L(z,w))$

f. P1: $(\forall x)(\forall y)((F(x) \wedge S(y)) \Rightarrow D(x,y))$
P2: $\sim (\exists x)(\exists y)(F(x) \wedge J(y) \wedge D(x,y))$

P3: $(\exists x)F(x)$

C :$(\forall y)(S(y) \Rightarrow \sim J(y))$

g. P1: $(\exists x)(P(x) \wedge (\forall y)(B(y) \Rightarrow R(x,y)))$
P2: $\sim(\exists x)(\exists y)(P(x) \wedge F(y) \wedge R(x,y))$
P3: $(\forall y)(F(y) \Rightarrow B(y))$

C :$\sim(\exists x)F(x)$

h. P1: $(\forall x)(A(x) \Rightarrow ((\forall y)(E(x,y) \Rightarrow H(y))))$
P2: $(\forall x)(T(x) \Rightarrow ((\exists y)(M(y) \wedge E(x,y)) \wedge (\exists y)(B(y) \wedge E(x,y))))$
P3: $(\forall x)(M(x) \Rightarrow H(x)) \wedge (\forall x)(B(x) \Rightarrow \sim H(x))$

C :$\sim(\exists x)(T(x) \wedge A(x))$

i. P1: $(\forall x)(P(x) \Rightarrow ((\exists y)(T(y) \wedge B(x,y)) \Rightarrow (\exists z)(G(z) \wedge R(x,z))))$
P2: $(\forall x)\ \sim G(x)$

C : $(\forall x)(\forall y)((P(x) \wedge T(y)) \Rightarrow \sim B(x,y))$

j. P1: $(\forall x)(R(x,x) \Rightarrow (\forall y)R(x,y))$
P2: $(\forall x)(\forall y)(R(x,y) \Rightarrow R(y,x))$
P3: $(\forall x)(\exists y)\ \sim R(y,x)$

C : $(\forall x)\ \sim R(x,x)$

2.4. [Lewis Carroll] Prove that each of the following conclusions follows from the given premise.

(a) My saucepans are the only things I have that are made of tin;
I find all your presents very useful;
None of my saucepans are of the slightest use.
Conclusion: Your presents to me are not made of tin.

(b) All the old articles in this cupboard are cracked;
No jug in this cupboard is new;
Nothing in this cupboard, that is cracked, will hold water.
Conclusion: No jug in this cupboard will hold water.

(c) Nobody who really appreciate Beethoven fails to keep silence while the *Moonlight-Sonata* is being played;
Guinea-pigs are hopelessly ingorant of music;
No one who is hopelessly ignorant of music ever keeps silence while the *Moonlight-Sonata* is being played.
Conclusion: Guinea-pigs never really appreciate Beethoven.

(d) Things sold in the street are of no great value;
Nothing but rubbish can be had for a song;
Eggs of the Great Auk are very valuable;
It is only what is sold in the street that is really rubbish.
Conclusion: An egg of the Great Auk is not to be had for a song.

(e) The only books in this library, that I do not recommend for reading are unhealthy in tone;
The bounded books are all well-written;
All the romances are healthy in tone;
I do not recommend you to read any of the unbounded books.

Conclusion: All the romances in this library are well-written.

(f) No interesting poems are unpopular among people of real taste;
No modern poetry is free from affection;
All your poems are on the subject of soap-bubbles;
No affected poetry is popular among people of real taste;
No ancient poem is on the subject of soap-bubbles.
Conclusion: All your poems are uninteresting.

(g) All writers, who understand human nature, are clever;
No one is a true poet unless he can stir the hearts of men;
Shakespeare wrote "*Hamlet*;"
No writers, who does not understand human nature, can stir the hearts of men;
None but a true poet could have written "*Hamlet*."
Conclusion: Shakespeare was clever.

(h) Animals, that do not kick, are always unexcitable;
Donkeys have no horns;
A buffalo can always toss one over a gate;
No animals that kick are easy to swallow;
No hornless animal can toss one over a gate;
All animals are excitable, except buffaloes.
Conclusion: Donkeys are not easy to swallow.

(i) No husband, who is always giving his wife new dresses, can be a cross-grained man;
A methodical husband always come home for his tea;
No one, who hangs up his hat on the gas-jet, can be a man that is kept in proper order by his wife;
A good husband is always giving his wife new dresses;
No husband can fail to be cross-grained, if his wife does not keep him in proper order;
An unmethodical husband always hangs up his hat on the gas-jet.
Conclusion: A good husband always come home for his tea.

(j) Animals are always mortally offened if I fail to notice them;
The only animals that belong to me are in that field;
No animal can guess a conundrum, unless it has been properly trained in a Board-School;
None of the animals in that field are badgers;
When an animal is mortally offened, it always rushes about wildly and howls;
I never notice any animal, unless it belongs to me;
No animal, that has been properly trained in a Board-School, ever rushes about wildly and howls.
Conclusion: No badger can guess a conundrum.

2.5. Prove that each of the following arguments is valid.

(a) People who consume neither meat nor fish are vegetarians.
Vegans consume no animal food or milk products.
Meat is animal food
Fish food is animal food.
Vegans are people.
Hence, vegans are vegetarians.

(b) A good research paper is a paper that contains outstanding results or is published in a tier-1 journal, and only such papers are good research papers.
Research papers containing outstanding results and research papers published in tier-1 journals are publications.
Hence, good research papers are publications.